THESIS REVISION GUIDE
Sections marked 01.11

Concrete remaining tasks

This checklist covers the three sections marked `01.11: Polish!` in Chapter 3. Line numbers refer to `5_bandwidth_selection.tex`. Sections marked 02.11 and citation cleanup are deferred. Items focus on mathematical accuracy and useful shortening.

3.1 KDE preliminaries

☐ **Make the setup and empirical convolution precise.** State that the observations are i.i.d. from f and that $h > 0$. Write the empirical convolution as $\int K_h(x-y)\,dP_n(y)$, rather than treating the discrete measure P_n as a density. [source lines 9–13]

☐ **Clarify the cross-validation target.** Explain that, for fixed h, the leave-one-out criterion is unbiased for $\mathrm{MISE}(h) - \int f^2$; distinguish this from the sample-dependent ISE. State the independence assumption used. [source lines 33–48] See Hang et al., Section 5.1.

☐ **Fix or remove the two convergence-rate claims.** The $O(n^{-1})$ ISE-gap claim and the $n^{-5/14}$ plug-in claim need precise assumptions, error quantities and stochastic convergence notation. The latter is stated as a *relative* bandwidth rate in Sheather–Jones, Eq. (10). If these details are not used later, omit the claims. [source lines 50, 115]

☐ **State the assumptions for MISE and AMISE.** Give the density/kernel regularity and bandwidth conditions needed for the L^2 calculations, bias expansion and oracle comparison; phrase $h_n \to 0$ and $nh_n \to \infty$ as sufficient under those assumptions. [source lines 54–100] See Hansen, Section 2.2, Theorem 2.1 and Chacón et al..

☐ **Align the selector discussion with the experiments.** The opening says plug-in and cross-validation methods are used for comparison. Ensure that matches the reported experiments. If they are background only, say so and shorten the detailed Sheather–Jones pilot derivation to the idea and resulting selector; otherwise verify its formulas. [source lines 23, 102–115; 551 onward]

3.3.1 Divergence matrices and the Perron vector

☐ **Repair the acute-chain lemma proof.** The diagonal entries of D are zero, so use positivity off the diagonal and irreducibility. The acute-chain assumption gives $\langle P_i, Q_i \rangle \geq 0$, not necessarily > 0. The last displayed maximum also needs squared norms. Supply the additional argument needed for the stated *strict* exclusion of interior maximizers, or weaken the conclusion to what the proof establishes. [source lines 319–380]

☐ **Shorten the same proof.** Replace the long expansion of μ by the one-line identity $\mu = x_1 + \sum_k a_k(x_{k+1} - x_k)$, then retain the essential comparison. Once the proof is complete, remove or fill the empty appendix subsection for the same lemma. [source lines 341–353; appendix lines 225–229]

☐ **Fix the Gaussian acute-chain calculation.** Explicitly use continuity of P_n and $P_n(0) = 1$. The Fourier-domain integrand in the positivity sentence must contain $b_k b_l$, as in the displayed equation, rather than $\delta_k \delta_l$. State the positive, strictly ordered bandwidth assumptions in the proposition. [source lines 387–405]

3.3.2 Similarity transformations

☐ **Repair the theorem's boundary assumption.** For completely monotone ϕ, require a finite continuous extension to 0 when applying it to $D(x, x) = 0$. In the Laplace representation include a possible atom at $t = 0$; this also covers constant functions. Nonconstant ϕ and distinct embedded points give the stated strict result. [source lines 419–449] See Guella, Theorem 3.1 and discussion following it.

☐ **Shorten the Schoenberg discussion.** Once the assumptions are correct, cite the existing theorem for the general equivalence and keep only the derivation or examples needed for the thesis. The Gaussian-kernel case is the one used in the experiments. [source lines 425–466]

☐ **Tidy the transformation's scope.** State $\tau > 0$ in the entropy minimization and that $S \to I$ as $\tau \downarrow 0$ requires $D_{ij} > 0$ for $i \neq j$. Keep the distinction between positive definiteness and entrywise positivity when explaining Perron–Frobenius. [source lines 468–483]

After the substantive revisions, compile the thesis and check these sections and cross-references. The linked sources support the review; bibliography cleanup remains deferred.